

# PRODUCT CIRCUIT DIAGRAM

A SYSTEMS VIEW OF HOW VALUE FLOWS THROUGH A PRODUCT — REVEALING FRICTION, MISMATCH, AND OPPORTUNITY

*“Utilities are sort of impedance-matched with the government. If you look at a utility, they are heavily regulated by public utility commissions. So they’re sort of impedance-matched with the government—both literally and figuratively” —Elon Musk*

## 🧠 CORE ANALOGY

| Electrical Concept   | Symbol | Product/System Equivalent                    |
|----------------------|--------|----------------------------------------------|
| Voltage (V)          | ⚡⚡⚡    | Strength of User Intent / Business Demand    |
| Current (I)          | —      | Flow of Value / progress                     |
| Resistance (R)       | —W—    | Friction (UX, process, org)                  |
| Impedance / Mismatch | └┐     | System Misalignment (handoffs, formats etc.) |
| Capacitor / Delay    | — — —  | Waiting / Buffering (queues, delays)         |
| Amplifier            | ▶      | Simplify / Automate / AI                     |
| Blackbox             | ■      | Stable, trusted subsystem                    |
| Whitebox             | □      | Problem area to inspect                      |
| Measurement          | 🕒      | Outcome (time, cost, success rate)           |

## ⚡ CREATING VALUE ⚡

Outcome Quality ∝ (Flow - Resistance - Mismatch - Delay) + Amplification

## THE UNLOCK 🔒

Instead of asking:

“What feature should we build?”

Ask:

“Where is energy being lost in the system?”